EXP 13	Case Study on Limitations of Low-Code Platforms and Proposing Hybrid Solutions

Aim:

To study the limitations of low-code platforms using the Hotel Room Booking application built in Zoho Creator and to propose hybrid (low-code + pro-code) solutions.

Theory:

Low-code platforms build apps quickly but have limits in customization, performance, integration and cost. A hybrid solution uses low-code for common parts and pro-code for special parts.

Case Study: The Hotel Room Booking application manages rooms and bookings, validates dates and capacity, updates room status, sends confirmation emails and shows a command center dashboard. The following limitations apply:

- **1. Limited UI design:** form and page look can be changed only within given options.
- **2. Plan limits:** trial plan limits users, apps and environments.
- **3. Script limits:** Deluge runs only inside Zoho with execution limits.
- **4. Vendor lock-in:** app and scripts work only on the Zoho platform.
- **5. Large data:** complex reports may slow down with many records.
- **6. Complex integrations:** payment gateways and channel managers need extra code.

Proposed Hybrid Solutions:

Limitation	Hybrid Solution
Limited UI design	Custom HTML/JS widgets and snippets
Plan limits	Upgrade plan or use fewer roles
Script limits	Custom functions / Zoho Catalyst
Vendor lock-in	Regular export and REST APIs
Large data	Move heavy processing to Catalyst functions
Complex integrations	Catalyst / Node.js functions for payments and OTA sync

Procedure:

1. Study the Hotel Room Booking application built in Zoho Creator.
2. List the limitations observed while building forms, workflows, pages and deployment.
3. Identify a pro-code or external service that can overcome each limitation.
4. Prepare the hybrid solution table and record observations.

Observation:

Feature	Low-Code Part	Pro-Code Part
Booking forms	Form builder	-
Validations	Deluge workflow	-
Custom UI panels	Page builder	HTML snippet / widget
Payments	-	Catalyst function + gateway API

Result:

The limitations of low-code were studied using the Hotel Room Booking application, and hybrid solutions combining Zoho Creator with custom widgets, functions and Zoho Catalyst were proposed.